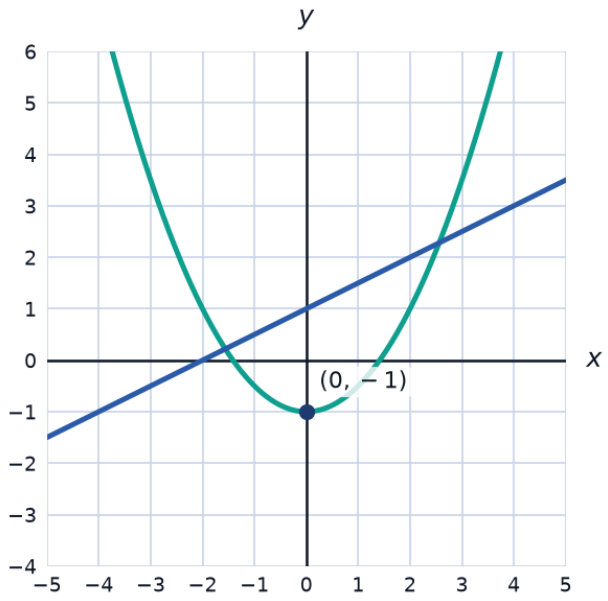
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A bag contains 8 red marbles, 6 green marbles, and 10 blue marbles. If 1 marble is chosen at random from the bag, what is the probability that the marble is NOT red?

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, the line segment with endpoints $A(1, -2)$ and $B(5, 6)$ is perpendicularly bisected by line ℓ. If the point $(9, k)$ lies on line ℓ, what is the value of k?



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The function $f(x) = a\sin(bx) + c$ has a maximum value of 9, a minimum value of -1, and a period of $\frac{\pi}{2}$. If $a > 0$ and $b > 0$, what is the value of $a + b + c$?

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, what is the distance between the points $(-1, 2)$ and $(3, 5)$?

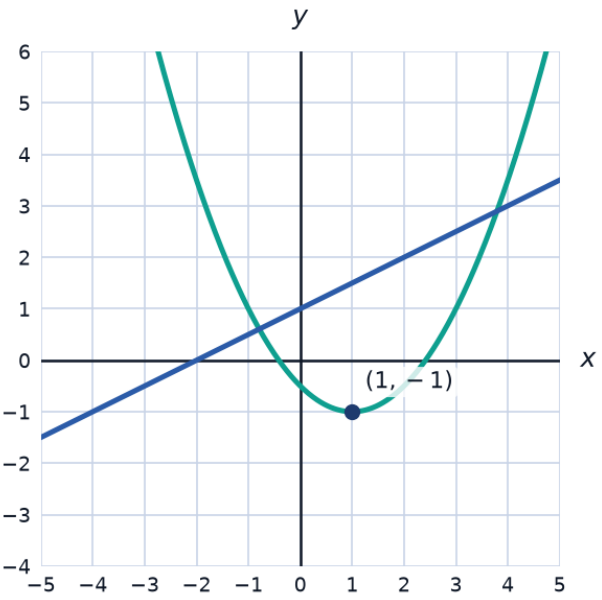
5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, vector $\mathbf{u} = \langle 2, -5 \rangle$ and vector $\mathbf{v} = \langle k, 3 \rangle$, where $k > 0$. If the magnitude of the vector sum $\mathbf{u} + \mathbf{v}$ is $\sqrt{29}$, what is the value of k?

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6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given the matrices $A = [2, -3; 1, 4]$ and $B = [-5, 1; 3, -2]$, what is the matrix $2A + B$?



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, what are the coordinates of the midpoint of a line segment with endpoints $(-3, 8)$ and $(5, -2)$?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

If $g(x) = 2x^2 - 5x + 1$, what is the value of $g(-3)$?

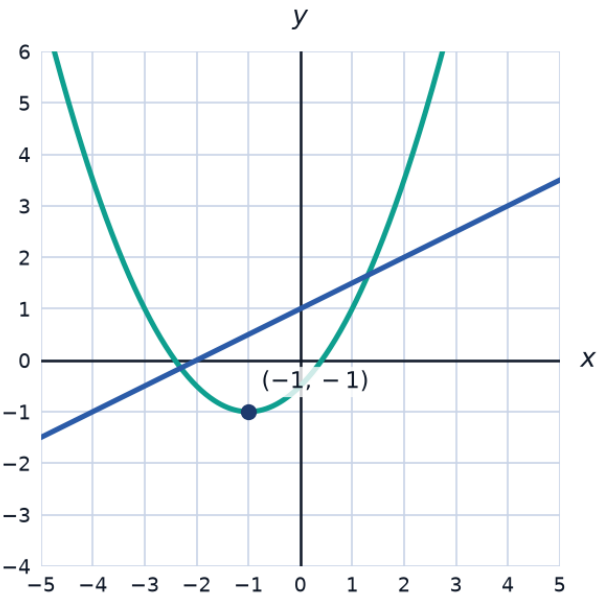
9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the standard (x, y) coordinate plane, what is the midpoint of the line segment with endpoints $(3, -4)$ and $(-5, 8)$?

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10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A fair coin is tossed twice. Find $P(\text{two heads})$.



11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A recipe is scaled from 2 to 5 servings. How many cups replace 4 cups?

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A square has area 36. Find its perimeter.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A right triangle has opposite 3 and hypotenuse 5. Find $\sin\theta$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A line passes through $(1, 2)$ with slope 3. Find y when $x = 4$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Functions & Graphs

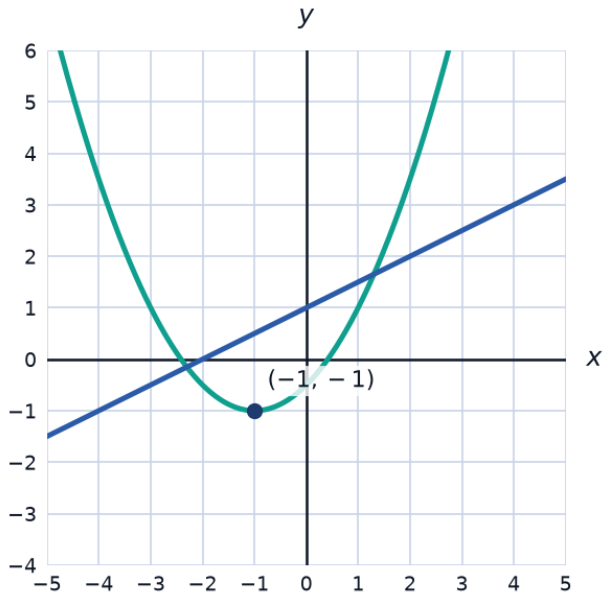
A random variable X has outcomes 7, 15, 19, 26 with

$P(X = 7) = \frac{3}{12}$, $P(X = 15) = \frac{5}{12}$, $P(X = 19) = \frac{2}{12}$, $P(X = 26) = \frac{2}{12}$. Find $E(X)$.

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16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Coordinate Geometry

A random variable X has outcomes -5, -2, -1, 0 with $P(X=-5)=\frac{2}{10}, P(X=-2)=\frac{2}{10}, P(X=-1)=\frac{4}{10}, P(X=0)=\frac{2}{10}$. Find $E(X)$.



17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Plane Geometry

For $f(x)=x^2$, find the average rate of change on $[1, 3]$.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Trigonometry

If $\sin\theta=-\frac{9}{41}$ and θ is in Quadrant IV, find $\cos\theta$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Statistics & Probability

Split 30 in the ratio $2:3$. Find the smaller part.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

A USD 80 item is marked down 20%. Find the sale price.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the linear equation for x: $\frac{2x-3}{4} - \frac{x+1}{3} = \frac{5}{12}$.

a. 7

b. 8

c. 9

d. 11

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $5(3x-2)+4 = 2(6x+7)-3$, what is the value of x?

a. 5

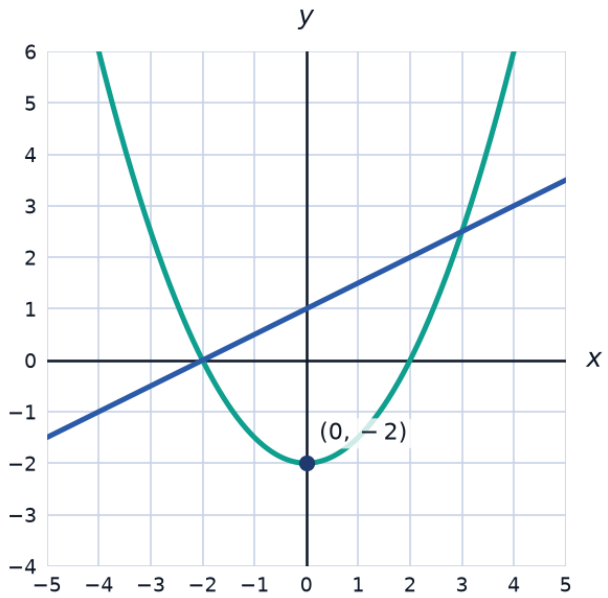
b. 6

c. 7

d. 8

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For what value of k does the equation $4(2x-5)+k = 8x-13$ have infinitely many solutions?



a. 7

b. 13

c. 20

d. −7

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $3x-2y = 19$ and $2x+5y = 0$, what is the value of $x+y$?

a. 3

b. 5

c. 7

d. 9

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the complete solution set for the compound inequality $-7 \leq 3x + 2 < 11$?

a. $-3 \leq x < 3$

b. $-5 \leq x < 4$

c. $-3 < x \leq 3$

d. $x \geq -3$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A company sells widgets for 14 each with fixed costs of $1,800$ and variable costs of 5 per widget. How many widgets must be sold to make a profit of at least 900?

a. 200

b. 250

c. 300

d. 350

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What are the solutions to the quadratic equation $2x^2 - 7x - 15 = 0$?

a. $x = 5$ and $x = -\frac{3}{2}$

b. $x = -5$ and $x = \frac{3}{2}$

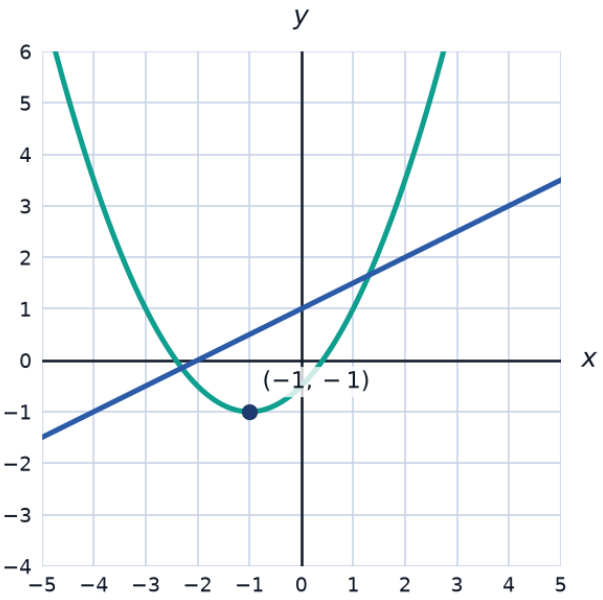
c. $x = 3$ and $x = -\frac{5}{2}$

d. $x = 5$ and $x = 3$

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the sum of the solutions of the equation $3x^2 - 12x + 5 = 0$?



a. -4

b. 4

c. $\frac{5}{3}$

d. 12

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of c will the quadratic equation $x^2 - 10x + c = 0$ have exactly one distinct real solution?

a. 5

b. 10

c. 25

d. 100

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Simplify the rational expression: $\frac{x^2 - 9}{2x^2 + 5x - 3}$ for all $x \neq \frac{1}{2}, -3$.

a. $\frac{x - 3}{2x - 1}$

b. $\frac{x + 3}{2x + 1}$

c. $\frac{x - 3}{2x + 1}$

d. $\frac{1}{2x - 1}$

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve the radical equation for all valid real values: $\sqrt{3x+13} = x+3$.

a. $x = 1$ only

b. $x = -4$ only

c. $x = 1$ and $x = -4$

d. No real solutions

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $\frac{1}{a} + \frac{1}{b} = \frac{5}{12}$ and $ab = 24$, what is the value of $a + b$?

a. 5

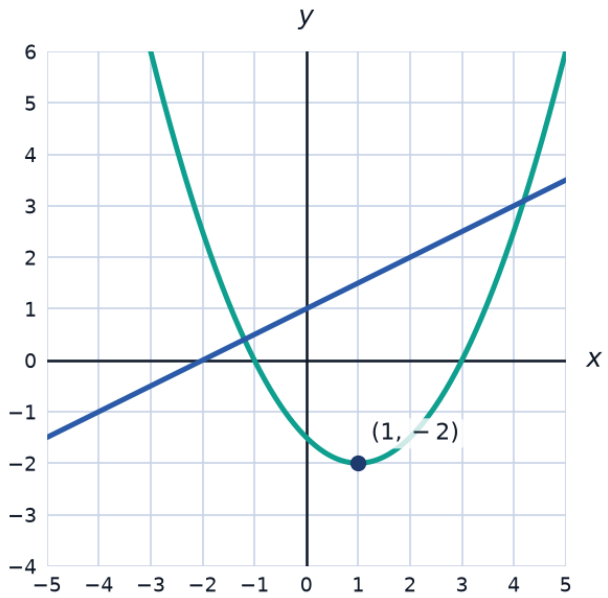
b. 10

c. 12

d. 15

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use the quadratic formula to solve $x^2 + 13x - 6 = 0$. What is the sum of the roots?



a. -17

b. -10

c. -16

d. -13

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use the quadratic formula to solve $x^2 - 11x - 10 = 0$. What is the sum of the roots?

a. 11

b. 5

c. 17

d. 9

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35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use the quadratic formula to solve $x^2 - 11x - 13 = 0$. What is the sum of the roots?

a. 15

b. 5

c. 8

d. 11

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use the quadratic formula to solve $2x^2 + 12x + 14 = 0$. What is the sum of the roots?

a. -2

b. -6

c. -1

d. -9

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use the quadratic formula to solve $3x^2 + 21x - 7 = 0$. What is the sum of the roots?

a. -6

b. -1

c. -7

d. -2

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

Use the quadratic formula to solve $3x^2 + 9x + 6 = 0$. What is the sum of the roots?

a. 0

b. -9

c. -3

d. 3

39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

Use the quadratic formula to solve $2x^2 - 6x - 8 = 0$. What is the sum of the roots?

a. 7

b. -3

c. 5

d. 3

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40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

Use the quadratic formula to solve $x^2 - 3x - 5 = 0$. What is the sum of the roots?

a. 8

b. 3

c. -3

d. 6

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

Use the quadratic formula to solve $x^2 - 3x - 13 = 0$. What is the sum of the roots?

a. 2

b. -3

c. 3

d. 5

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Functions & Graphs

Use the quadratic formula to solve $x^2 - 8x + 12 = 0$. What is the sum of the roots?

a. 3

b. 10

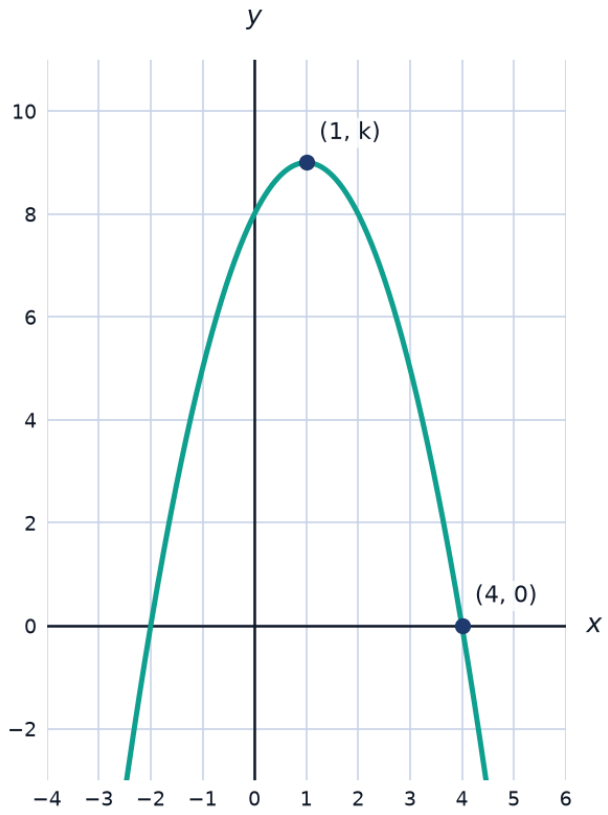
c. 8

d. 9

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

The graph of the quadratic function $f(x) = -x^2 + bx + c$ is shown in the standard (x, y) coordinate plane below. The vertex of the parabola is at $(1, k)$, and the graph passes through the point $(4, 0)$. What is the value of k?



a. 6

b. 8

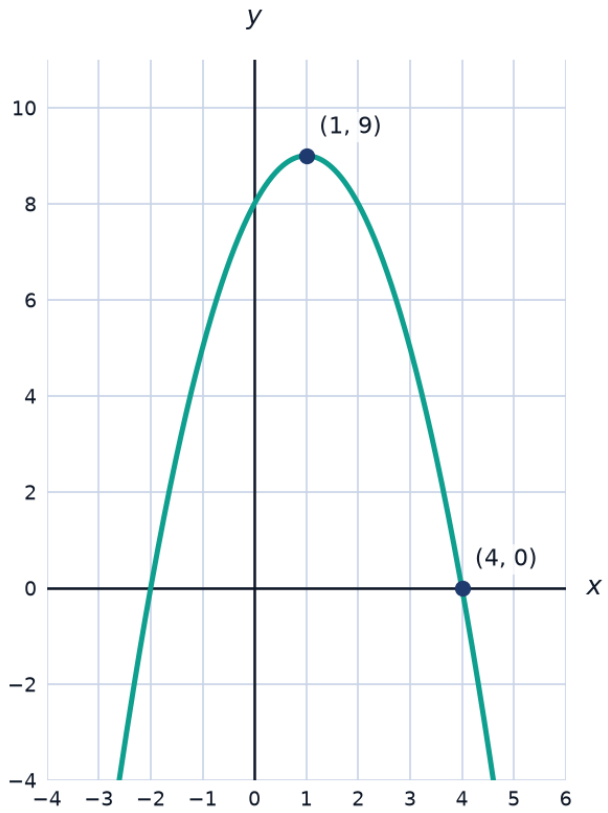
c. 9

d. 12

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Plane Geometry

The graph of the quadratic function $f(x) = -x^2 + bx + c$ is shown in the standard (x, y) coordinate plane below. The vertex of the parabola is at $(1, 9)$, and the graph passes through the point $(4, 0)$. What is the value of $b - c$?



a. -10

b. -7

c. -6

d. 6

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

The quadratic function $f(x) = x^2 - 2kx + (3k + 10)$ has 2 distinct real zeros. If both zeros are strictly greater than 2, how many integer values of k are possible?

a. 7

b. 8

c. 9

d. 11

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